1. Zadatak

Za date konačne jednačine kretanja u Dekartovim koordinatama tačke M u metrima i vreme u sekundama odrediti: brzinu i ubrzanje tačke M u proizvoljnom vremenu, a potom tangencijalno i normalno ubrzanje tačke M u trenutkiu t=π/6.

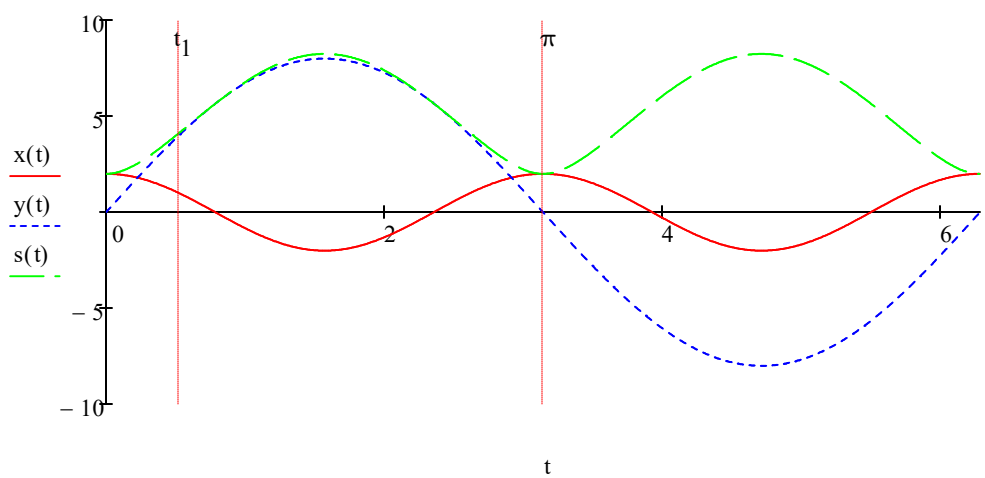
$$x(t) := 2\cdot\cos(2t)$$

$$y(t) := 8\cdot\sin(t)$$

Putanja tačke M: $s(t) := \sqrt{(x(t))^2 + (y(t))^2}$ $\qquad s(t) \rightarrow 2\cdot\sqrt{\cos(2\cdot t)^2 + 16\cdot\sin(t)^2}$

za trenutak vremena: $t_1 := \frac{\pi}{6}$ od $T := 2\pi$ $\qquad x(t_1) = 1\ \ m \qquad y(t_1) = 4\ \ m$

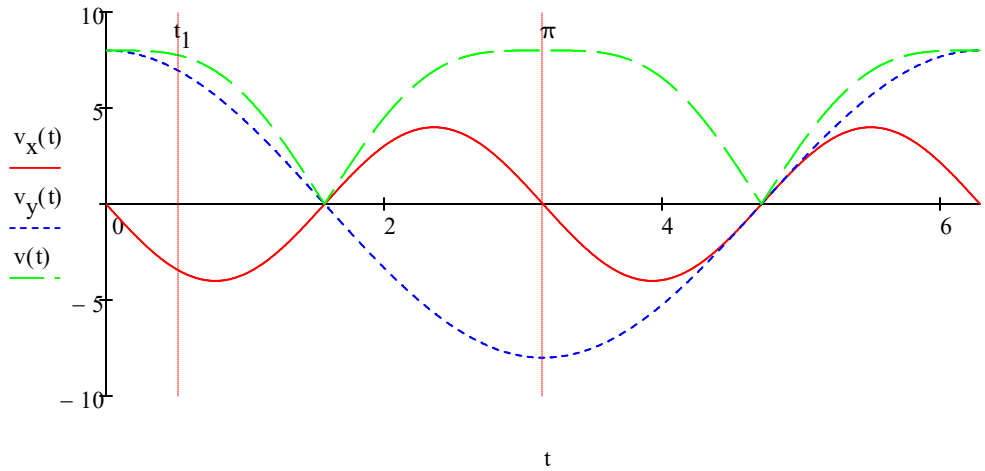
$$s(t_1) = 4.123\ \ m$$



Brzina tačke M: $v_x(t) := \frac{d}{dt}x(t)$ $\qquad v_x(t) \rightarrow -4\cdot\sin(2\cdot t)$

$$v_y(t) := \frac{d}{dt}y(t) \qquad v_y(t) \rightarrow 8\cdot\cos(t)$$

$$v(t) := \sqrt{(v_x(t))^2 + (v_y(t))^2} \qquad v(t) \rightarrow 4\cdot\sqrt{\sin(2\cdot t)^2 + 4\cdot\cos(t)^2}$$

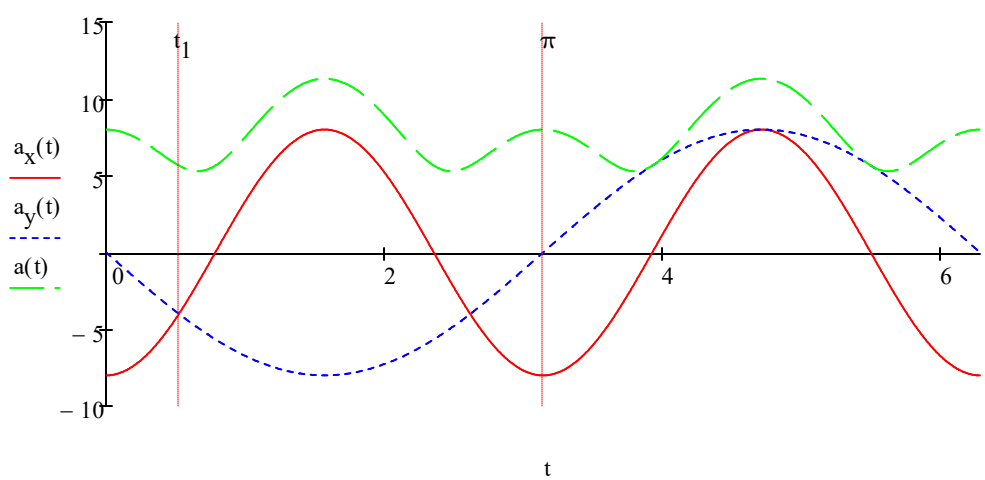


Ubrzanje tačke M:

$$a_x(t) := \frac{d}{dt} v_x(t) \qquad a_x(t) \rightarrow -8 \cdot \cos(2 \cdot t)$$

$$a_y(t) := \frac{d}{dt} v_y(t) \qquad a_y(t) \rightarrow -8 \cdot \sin(t)$$

$$a(t) := \sqrt{\left(a_x(t)\right)^2 + \left(a_y(t)\right)^2} \qquad a(t) \rightarrow 8 \cdot \sqrt{\cos(2 \cdot t)^2 + \sin(t)^2}$$



Ukupnog ubrzanja tačke M je:

$$a(t) = \sqrt{\left(a_T(t)\right)^2 + \left(a_N(t)\right)^2}$$

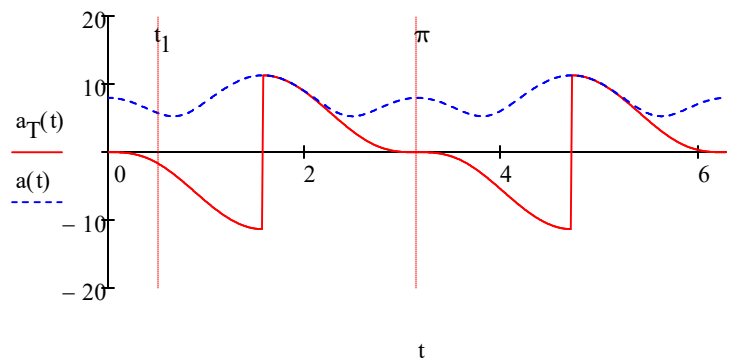
odavde najpre odredimo komponentu **tangencijalnog ubrzanja**:

$$a_T(t) := \frac{d}{dt}v(t)$$

$$a_T(t) \rightarrow \frac{8\cdot\cos(2\cdot t)\cdot\sin(2\cdot t) - 16\cdot\cos(t)\cdot\sin(t)}{\sqrt{\sin(2\cdot t)^2 + 4\cdot\cos(t)^2}}$$

$$a_T(t_1) \text{ simplify } \rightarrow -\frac{4\cdot\sqrt{5}}{5}$$

$$a_T(t_1) = -1.789$$

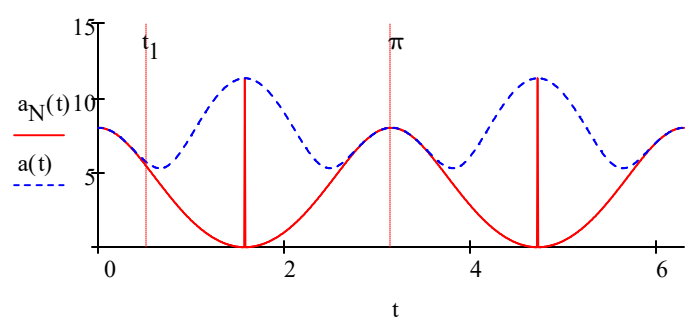


na kraju nalazimo komponentu **normalnog ubrzanja** da je:

$$a_N(t) := \sqrt{(a(t))^2 - \left(a_T(t)\right)^2}$$

$$a_N(t_1) \rightarrow \frac{12\cdot\sqrt{5}}{5}$$

$$a_N(t_1) = 5.367$$



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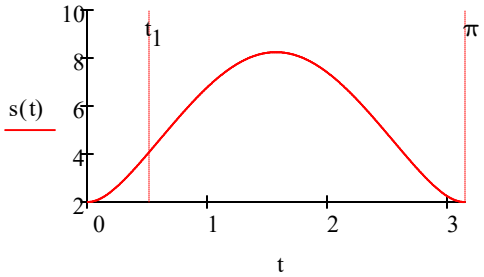
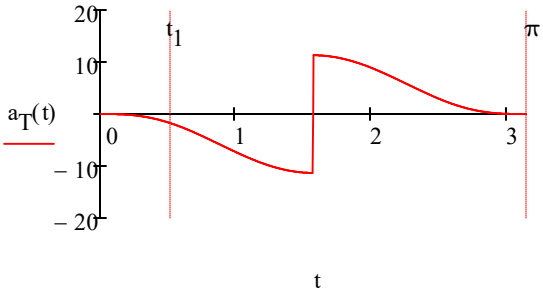
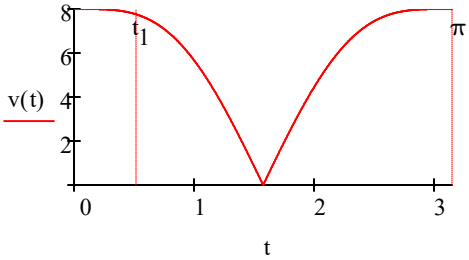
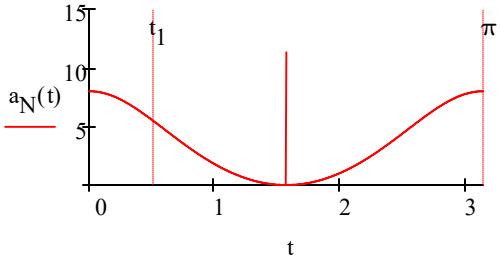
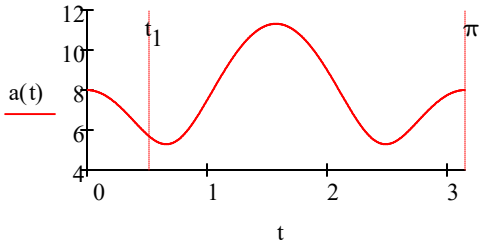
$T := \frac{T}{2}$

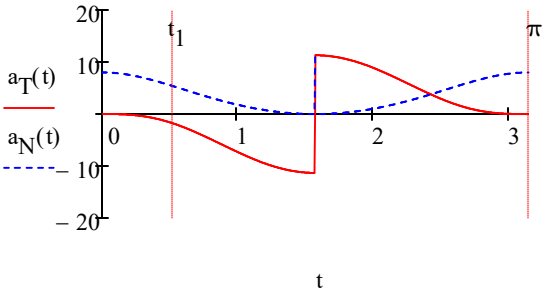


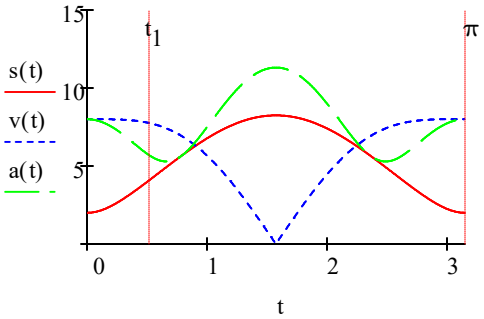


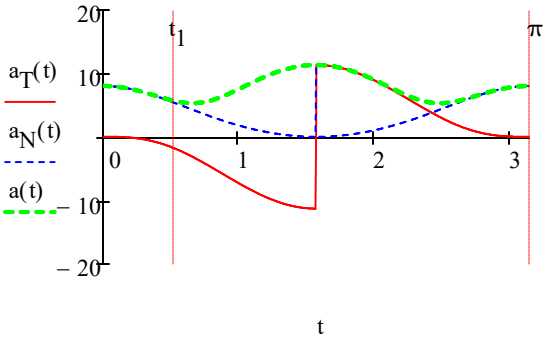




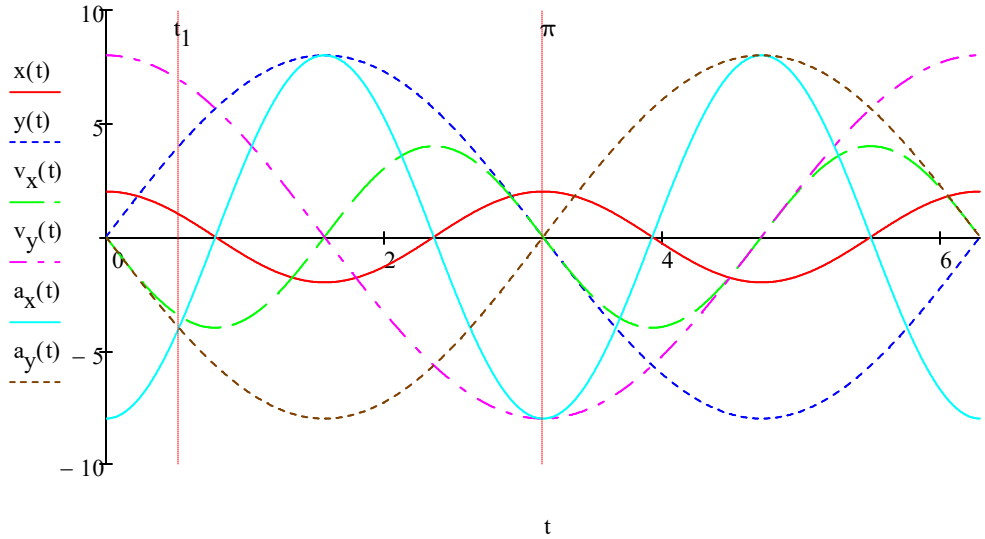


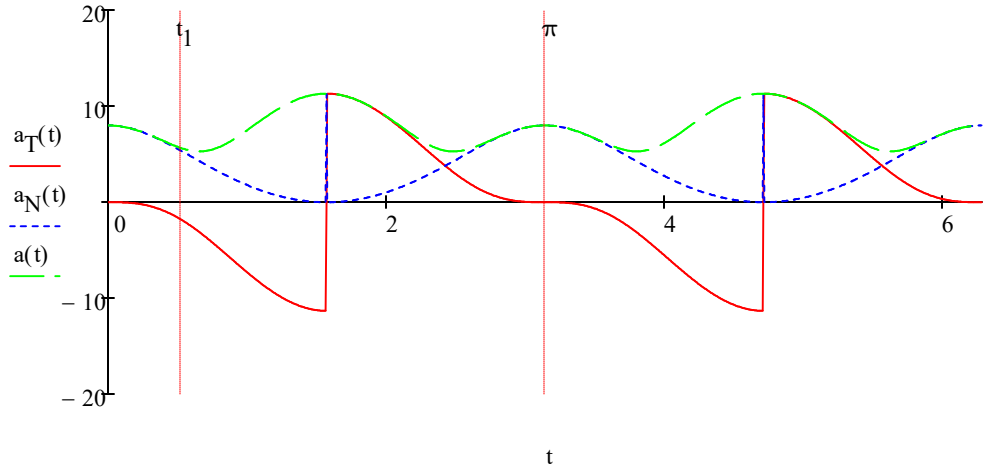


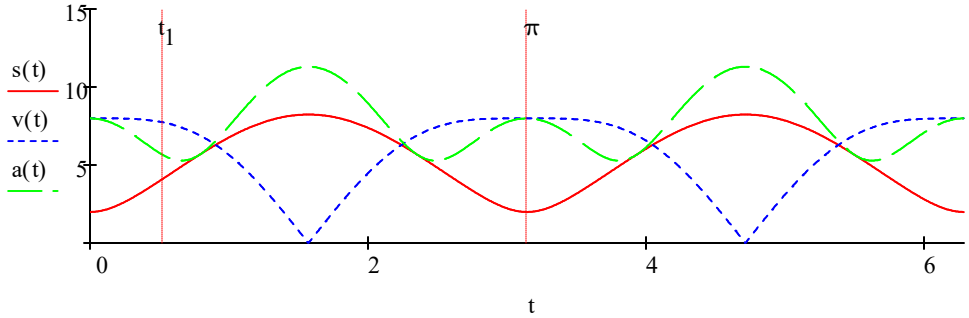




$T := 2T$

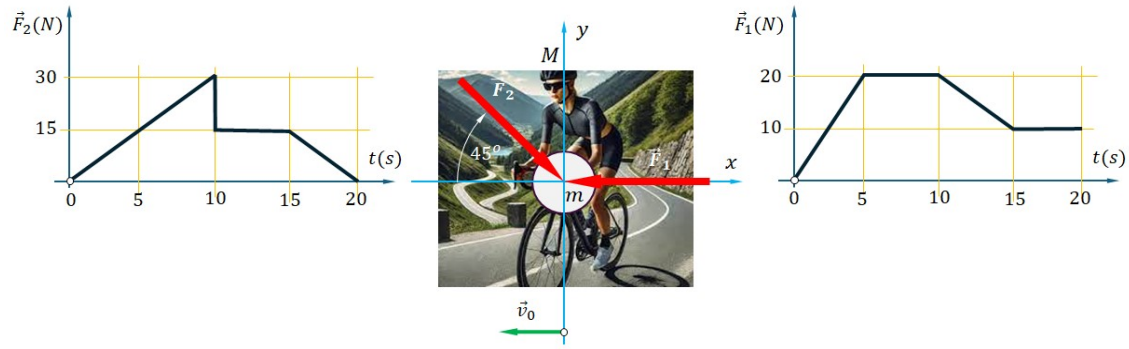

10
5
− 5
− 10
0
2
4
6
t1
π
x(t)
y(t)
vx(t)
vy(t)
ax(t)
ay(t)
t


20
10
− 10
− 20
0
2
4
6
t1
π
aT(t)
aN(t)
a(t)
t


15
10
5
0
2
4
6
t1
π
s(t)
v(t)
a(t)
t

2. Zadatak

Na osnovu date skice, primenom zakona o promeni količine kretanja sa zanemarivanjem težine materijalne tačke usled dejstva sila F1 i F2, sračunati: pravac vektora brzine tačke M, kao i njegov intenzitet u trenutku t=20s.



Sračunavanje pravaca i smerova impulsa sila jer su oni kolinearni sa silama:

$$I_1 = \int_{t_0}^{t_1} F_1\,dt \qquad I_1 := \frac{20\cdot 5}{2} + 20\cdot 5 + \frac{20+10}{2}\cdot 5 + 10\cdot 5 \qquad I_1 = 275$$

$$I_2 = \int_{t_0}^{t_1} F_2\,dt \qquad I_2 := \frac{30\cdot 10}{2} + 15\cdot 5 + \frac{15\cdot 5}{2} \qquad I_2 = 262.5$$

Primenom zakona o promeni količine kretanja nalazimo projekcije sila:

Given $\quad m := 100 \quad \alpha := 0 \quad v_{ox} := 10 \quad \beta := 45 \quad v_{oy} := 0$

x-pravac: $\quad m\cdot v_{1x} - m\cdot v_{ox} = -I_1\cdot\cos(\alpha\cdot deg) + I_2\cdot\cos(\beta\cdot deg)$

y-pravac: $\quad m\cdot v_{1y} - m\cdot v_{oy} = I_1\cdot\sin(\alpha\cdot deg) - I_2\cdot\sin(\beta\cdot deg)$

Rešenja sistema jednačina:

$$\mathrm{Find}(v_{1x}, v_{1y}) \rightarrow \begin{pmatrix} \dfrac{21\cdot\cos(45\cdot deg)}{8} + \dfrac{29}{4} \\ -\dfrac{21\cdot\sin(45\cdot deg)}{8} \end{pmatrix}$$

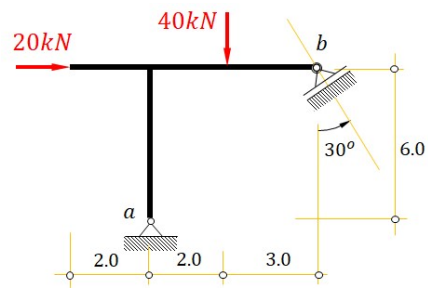
$$v_{1x} := \frac{21\cdot\cos(45\cdot deg)}{8} + \frac{29}{4} \qquad v_{1x} = 9.106$$

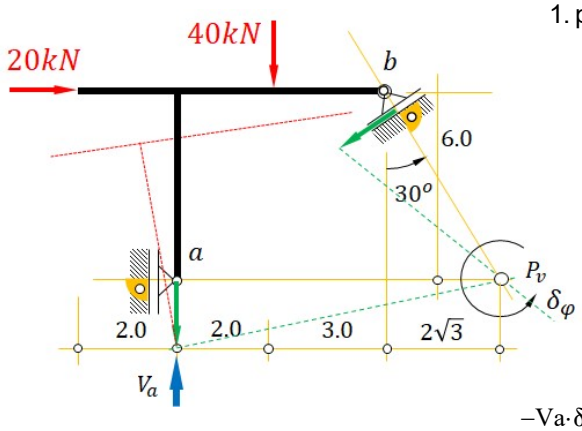
$$v_{1y} := -\frac{21\cdot\sin(45\cdot deg)}{8} \qquad v_{1y} = -1.856$$

Intenzitet brzine: $\quad v_M := \sqrt{(v_{1x})^2 + (v_{1y})^2} \qquad v_M = 9.293\ \frac{m}{s}$

3. zadatak

Za nosač sa opterećenjem prikazan na skici primenom opšte jednačine statike sračunati rekacije u osloncu a.





1. plan pomeranja:

$$\delta\phi := \frac{1}{\left(2 + 3 + 2\sqrt{3}\right)}$$

$$\delta\phi = 0.118$$

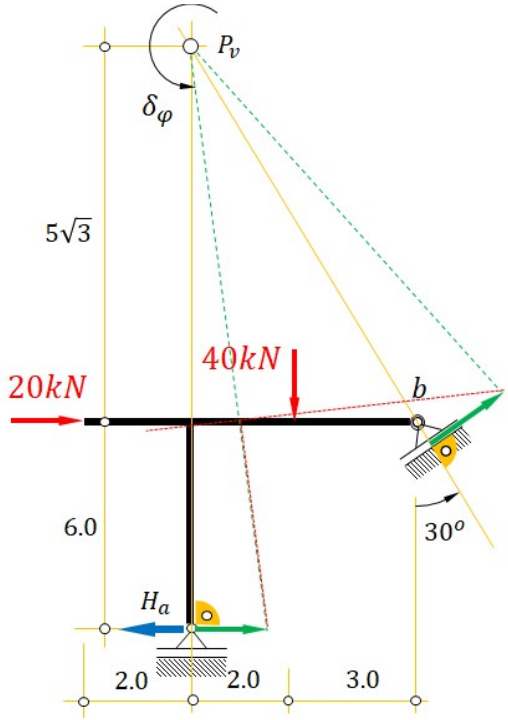
$$\delta A = 0$$

Given

$$-Va\cdot\delta\phi\cdot\left(2 + 3 + 2\cdot\sqrt{3}\right) - 20\cdot\delta\phi\cdot 6 + 40\cdot\delta\phi\cdot\left(3 + 2\sqrt{3}\right) = 0$$

$$\text{Find}(Va) \rightarrow \frac{80\cdot\sqrt{3}}{2\cdot\sqrt{3} + 5} \qquad Va := \frac{80\cdot\sqrt{3}}{2\cdot\sqrt{3} + 5}$$

$$Va = 16.371$$



2. plan pomeranja:

$$\delta\phi := \frac{1}{\left(6 + 5\sqrt{3}\right)}$$

$$\delta\phi = 0.068$$

$$\delta A = 0$$

Given

$$-Ha\cdot\delta\phi\cdot\left(6 + 5\cdot\sqrt{3}\right) + 20\cdot\delta\phi\cdot 5\sqrt{3} - 40\cdot\delta\phi\cdot 2 = 0$$

$$\text{Find}(Ha) \rightarrow \frac{20\cdot\left(5\cdot\sqrt{3} - 4\right)}{5\cdot\sqrt{3} + 6} \qquad Ha := \frac{20\cdot\left(5\cdot\sqrt{3} - 4\right)}{5\cdot\sqrt{3} + 6}$$

$$Ha = 6.358$$